

MAT 426, Class 20 Homework: Singular Value Decomposition
(Due Tuesday, April 28, 2026)

1. Find the singular value decomposition of the given matrix A .

(a) $A = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$ (b) $A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$ (c) $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$ (d) $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$

2. Show that if A is an $n \times n$ positive definite matrix, then an orthogonal decomposition $A = PDP^T$ is a singular value decomposition of A .
3. Let $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_n]$ and $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ where the $\mathbf{u}_i$ and $\mathbf{v}_i$ are as in the singular value decomposition of A . Show that

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T.$$